JavaScript is disabled on your browser.

compact1, compact2, compact3

java.math

Class MathContext

java.lang.Object
    java.math.MathContext

All Implemented Interfaces:

Serializable

```
public final class MathContext
extends Object
implements Serializable
```

Immutable objects which encapsulate the context settings which describe certain rules for numerical operators, such as those implemented by the `BigDecimal` class.

The base-independent settings are:

1. `precision`: the number of digits to be used for an operation; results are rounded to this precision
2. `roundingMode`: a `RoundingMode` object which specifies the algorithm to be used for rounding.

Since:

1.5

See Also:

BigDecimal, RoundingMode, Serialized Form

Field Summary

Fields

Modifier and Type	Field	Description

static MathContext	DECIMAL128	A MathContext object with a precision setting matching the IEEE 754R Decimal128 format, 34 digits, and a rounding mode of HALF_EVEN, the IEEE 754R default.
static MathContext	DECIMAL32	A MathContext object with a precision setting matching the IEEE 754R Decimal32 format, 7 digits, and a rounding mode of HALF_EVEN, the IEEE 754R default.
static MathContext	DECIMAL64	A MathContext object with a precision setting matching the IEEE 754R Decimal64 format, 16 digits, and a rounding mode of HALF_EVEN, the IEEE 754R default.
static MathContext	UNLIMITED	A MathContext object whose settings have the values required for unlimited precision arithmetic.

Constructor Summary

Constructors

Constructor	Description
MathContext(int setPrecision)	Constructs a new MathContext with the specified precision and the HALF_UP rounding mode.
MathContext(int setPrecision, RoundingMode setRoundingMode)	Constructs a new MathContext with a

	specified precision and rounding mode.
`MathContext(String val)`	Constructs a new `MathContext` from a string.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`boolean`	`equals(Object x)`	Compares this `MathContext` with the specified `Object` for equality.
`int`	`getPrecision()`	Returns the `precision` setting.
`RoundingMode`	`getRoundingMode()`	Returns the roundingMode setting.
`int`	`hashCode()`	Returns the hash code for this `MathContext`.
`String`	`toString()`	Returns the string representation of this `MathContext`.

Methods inherited from class java.lang.Object

`clone`, `finalize`, `getClass`, `notify`, `notifyAll`, `wait`, `wait`, `wait`

Field Detail

UNLIMITED

```
public static final MathContext UNLIMITED
```

A `MathContext` object whose settings have the values required for unlimited precision arithmetic. The values of the settings are: `precision=0 roundingMode=HALF_UP`

DECIMAL32

```
public static final MathContext DECIMAL32
```

A `MathContext` object with a precision setting matching the IEEE 754R Decimal32 format, 7 digits, and a rounding mode of `HALF_EVEN`, the IEEE 754R default.

DECIMAL64

```
public static final MathContext DECIMAL64
```

A `MathContext` object with a precision setting matching the IEEE 754R Decimal64 format, 16 digits, and a rounding mode of `HALF_EVEN`, the IEEE 754R default.

DECIMAL128

```
public static final MathContext DECIMAL128
```

A `MathContext` object with a precision setting matching the IEEE 754R Decimal128 format, 34 digits, and a rounding mode of `HALF_EVEN`, the IEEE 754R default.

Constructor Detail

MathContext

```
public MathContext(int setPrecision)
```

Constructs a new `MathContext` with the specified precision and the `HALF_UP` rounding mode.

Parameters:

`setPrecision` - The non-negative int precision setting.

Throws:

`IllegalArgumentException` - if the setPrecision parameter is less than zero.

MathContext

```
public MathContext(int setPrecision,
                   RoundingMode setRoundingMode)
```

Constructs a new `MathContext` with a specified precision and rounding mode.

Parameters:

`setPrecision - The non-negative int precision setting.`

`setRoundingMode - The rounding mode to use.`

Throws:

`IllegalArgumentException - if the setPrecision parameter is less than zero.`

`NullPointerException - if the rounding mode argument is null`

MathContext

```
public MathContext(String val)
```

Constructs a new `MathContext` from a string. The string must be in the same format as that produced by the `toString()` method.

An `IllegalArgumentException` is thrown if the precision section of the string is out of range (`< 0`) or the string is not in the format created by the `toString()` method.

Parameters:

`val - The string to be parsed`

Throws:

`IllegalArgumentException - if the precision section is out of range or of incorrect format`

`NullPointerException - if the argument is null`

Method Detail

getPrecision

```
public int getPrecision()
```

Returns the `precision` setting. This value is always non-negative.

Returns:

an int which is the value of the precision setting

getRoundingMode

public RoundingMode getRoundingMode()

Returns the roundingMode setting. This will be one of RoundingMode.CEILING, RoundingMode.DOWN, RoundingMode.FLOOR, RoundingMode.HALF_DOWN, RoundingMode.HALF_EVEN, RoundingMode.HALF_UP, RoundingMode.UNNECESSARY, or RoundingMode.UP.

Returns:

a RoundingMode object which is the value of the roundingMode setting

equals

public boolean equals(Object x)

Compares this MathContext with the specified Object for equality.

Overrides:

equals in class Object

Parameters:

x - Object to which this MathContext is to be compared.

Returns:

true if and only if the specified Object is a MathContext object which has exactly the same settings as this object

See Also:

Object.hashCode(), HashMap

hashCode

public int hashCode()

Returns the hash code for this MathContext.

Overrides:

hashCode in class Object

Returns:

hash code for this MathContext

See Also:

Object.equals(java.lang.Object),
System.identityHashCode(java.lang.Object)

toString

```
public String toString()
```

Returns the string representation of this `MathContext`. The `String` returned represents the settings of the `MathContext` object as two space-delimited words (separated by a single space character, `'\u0020'`, and with no leading or trailing white space), as follows:

1. The string `"precision="`, immediately followed by the value of the precision setting as a numeric string as if generated by the `Integer.toString` method.
2. The string `"roundingMode="`, immediately followed by the value of the `roundingMode` setting as a word. This word will be the same as the name of the corresponding public constant in the `RoundingMode` enum.

For example:

```
precision=9 roundingMode=HALF_UP
```

Additional words may be appended to the result of `toString` in the future if more properties are added to this class.

Overrides:

`toString` in class `Object`

Returns:

a String representing the context settings

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